

Selective Tool Retrieval for AI Agents

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Fresh queries	Candidate tools	Source domains	Training in this pilot
300	37,292	3 x 100 queries	None

Research question

Can agreement between a sparse retriever and a dense retriever identify queries that can skip cross-encoder reranking, while preserving retrieval quality across several ToolRet source domains?

Abstract

This study evaluates a fixed, zero-training disagreement gate on 300 fresh queries over the ToolRet web corpus. The gate reranks a query only when BM25 and base MiniLM retrieve different top-ranked tools. It uses 27.7% fewer reranker invocations than always-on reranking. The gated-minus-always nDCG@10 difference is +0.0033, with a paired query-bootstrap 95% interval [-0.0058, +0.0128]. These findings describe this pilot and its compute counts; no adaptive service latency or downstream agent success was measured.

Fixed experimental design

Select 100 queries each from APIGen, ToolBench, and ToolACE by ascending SHA256(seed + NUL + query ID). Normalize Unicode, case, and whitespace; remove duplicates and APIBank text overlaps before selection. Every positive label must refer to a tool in the corpus. Retrieval uses the query alone; label documents and instructions are excluded from model inputs.

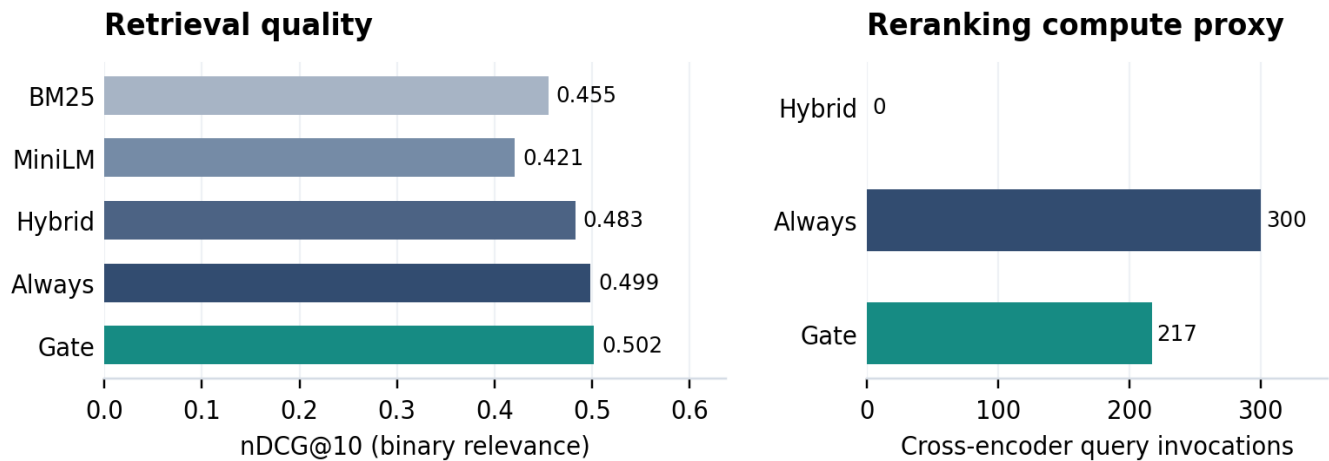
Component	Configuration
Dense retriever	sentence-transformers/all-MiniLM-L6-v2
Cross-encoder	cross-encoder/ms-marco-MiniLM-L-6-v2
Candidates / fusion	Top 100 per retriever; weighted RRF k=60, weights=[1.0, 1.0]
Gate / rerank depth	BM25 top-1 != MiniLM top-1; rerank top 20

Public pretrained models are pinned to revisions; neural sequence length is limited to 256 tokens. No weights, gate thresholds, or retrieval settings were trained or selected using this pilot. This is an independent extension, not a submitted paper or a faculty-supervised appointment.

Results: retrieval quality and compute

All quality metrics use binary relevance and cutoff 10. MRR@10 measures the reciprocal rank of the first positive result; Recall@10 averages the fraction of each query's positive labels retrieved; nDCG@10 rewards their ranking positions. All-label coverage requires every positive label to appear in the top 10.

Policy	MRR@10	nDCG@10	Recall@10	All-label coverage	CE calls	CE pairs
BM25	0.4715	0.4553	0.5614	0.4567	0	0
Base MiniLM	0.4387	0.4210	0.5330	0.4267	0	0
Hybrid RRF	0.4785	0.4828	0.6323	0.5267	0	0
Always rerank	0.5113	0.4986	0.6196	0.5100	300	6000
Disagreement gate	0.5180	0.5019	0.6173	0.5100	217	4340



The chart reports retrieval quality and cross-encoder invocation counts. The right panel is a compute proxy, not measured response latency.

Paired uncertainty

Comparison	nDCG difference [95% CI]	MRR difference [95% CI]
Always - hybrid	+0.0158 [-0.0090, +0.0410]	+0.0328 [-0.0015, +0.0668]
Gate - hybrid	+0.0191 [-0.0041, +0.0423]	+0.0395 [+0.0078, +0.0713]
Gate - always	+0.0033 [-0.0058, +0.0128]	+0.0067 [-0.0045, +0.0191]

Intervals use 20,000 paired query-bootstrap resamples, seed 20261003, and percentile bounds. They assume exchangeable query observations and do not account for dependency within API families. They are exploratory, not multiplicity-adjusted.

The primary gate-versus-always interval includes zero. This pilot establishes neither superiority nor formal noninferiority; the interval does not prove equivalent quality.

An equal-invocation random policy across 20 fixed seeds achieves mean nDCG@10 0.4932 (seed range 0.4816-0.5018), using 217 reranker calls per seed. This seed range describes random-policy variation; it is not a confidence interval.

Domain behavior and evidence boundaries

Source-domain breakdown

Source	Queries	Gate calls	Hybrid nDCG	Always nDCG	Gate nDCG	Gate MRR
apigen	100	69	0.6120	0.6868	0.6818	0.6937
toolace	100	62	0.5588	0.5551	0.5597	0.5551
toolbench	100	86	0.2775	0.2539	0.2642	0.3052

Domain regressions matter: toolbench gated-minus-hybrid nDCG@10 is -0.0134. Aggregate performance does not imply a benefit for every source.

Source datasets are not API families: upstream metadata contains no structured API-family identity. Means are over queries; source groups have equal sample sizes.

Compute savings and measured components

The gate saves 27.7% of reranker invocations and 27.7% of scoring pairs versus always reranking. The paired-evaluation cache computes always-on rankings for every query.

A separate selective-runtime check actually made 217 CE calls over 4,340 pairs and skipped 83 calls. All 300 resulting rankings matched the cache. It reused cached first-stage rankings and does not measure full request latency.

Independent recomputation passed 21 integrity checks and matched all recorded metrics within tolerance.

Measured component	Mean ms	p50 ms	p95 ms
BM25 search	233.47	221.09	376.70
Dense encode + exact search	31.15	28.92	45.98
RRF fusion	0.21	0.19	0.27
CE scoring + prefix sorting	635.55	628.12	863.48

CPU, 4 PyTorch threads; warmed models; one measurement per query. Timings exclude setup, encoding/index build, network, concurrency, and service overhead. Counterfactual offline sums omit routing overhead and do not measure adaptive request latency.

Historical result audit

The saved 16-query APIBank CSV exposed old README discrepancies when MRR was recomputed. These are historical audit findings, not new model evaluations.

Historical system	Old README MRR	Saved-rank MRR	Matches rounded value
BM25	0.132	0.146065	No
Hard-negative MiniLM	0.499	0.521205	No
Hybrid retrieval	0.515	0.514667	Yes
Hybrid + reranker	0.473	0.473000	Yes

Historical MRR has no cutoff at 10 and refers to unavailable trained checkpoints, so it is not comparable to this pilot's base-model MRR@10. Saved first-relevant ranks cannot recover Recall/nDCG, full rankings, or original run provenance. Inspected test data were preserved and excluded from the fresh pilot.

Reproducibility, limitations, and context

What the artifact supports

The artifact supports a preliminary study with pinned data/models, deterministic selection, ranking caches, a fixed label-independent gate, paired uncertainty, source breakdowns, and a historical audit. It does not establish technique priority, benchmark-wide superiority, production speedup, or downstream task completion.

Limitations and next experiment

The 300-query pilot does not cover the full benchmark. Query dependence and incomplete relevance annotations can affect estimates; public-model pretraining overlap is unknown. All-label coverage does not establish mandatory-tool coverage. No model was trained here, and source domains do not define API-family holdouts.

Next, establish API-family metadata independently of labels, reserve separate development/evaluation groups, expand the cohort, and measure adaptive request latency repeatedly. Fit future learned gates or thresholds outside today's pilot.

Reproduce from the repository root

```
D=data/research_20261003; R=results/research_20261003
python scripts/prepare_research_pilot.py
PYTHONPATH=src python scripts/generate_research_cache.py \
  --corpus "$D/corpus.jsonl" --queries "$D/queries.jsonl" \
  --data-manifest "$D/manifest.json" \
  --protocol configs/selective_pilot_20261003.json \
  --output-dir "$R/pilot_cache" --embedding-dir "$D/embeddings"
PYTHONPATH=src python scripts/evaluate_selective_reranking.py \
  --cache "$R/pilot_cache/rankings.jsonl" --corpus "$D/corpus.jsonl" \
  --queries "$D/queries.jsonl" --manifest "$R/pilot_cache/manifest.json" \
  --exclude-queries results/test_failure_analysis.csv \
  --bootstrap-resamples 20000 --seed 20261003 \
  --output-dir "$R/pilot_eval"
```

Use Python 3.12 with CPU PyTorch, sentence-transformers, and pyarrow; exact executed package versions are in the cache manifest. Reruns require new output directories. Evidence JSON files retain complete hashes and timing provenance.

Pinned model revisions

```
sentence-transformers/all-MiniLM-L6-v2
1110a243fdf4706b3f48f1d95db1a4f5529b4d41

cross-encoder/ms-marco-MiniLM-L-6-v2
233902d25c440f23af6f7d6e94d2946bac0bee0a
```

Primary sources and related work

[1] Shi et al. (2025). Retrieval Models Aren't Tool-Savvy: Benchmarking Tool Retrieval for Large Language Models. Findings of ACL. ToolRet supplies the benchmark; this pilot uses only its web corpus.
<https://aclanthology.org/2025.findings-acl.1258/>

[2] Zheng et al. (2024). ToolRerank: Adaptive and Hierarchy-Aware Reranking for Tool Retrieval. LREC-COLING. Adaptive tool reranking predates this study.
<https://aclanthology.org/2024.lrec-main.1413/>

[3] Lookahead-R: Budget-Aware Tool Retrieval via Execution-Centric Planning. arXiv:2609.35811 (2026), preprint. Related budget-aware tool retrieval work.
<https://arxiv.org/abs/2609.35811>

Data: mangopy/ToolRet-Queries and mangopy/ToolRet-Tools. Both dataset revisions and downloaded-file hashes are in the preparation manifest.

Code: <https://github.com/michaelbawuah/toolret-hybrid-retrieval>